

회귀분석

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Data science summer institute

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Simple Linear Regression

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Agenda

1. Covariance^{표본공분산} and Correlation Coefficient^{표본상관계수}
2. The Simple Linear Regression Model^{단순선형회귀모형}
3. Parameter Estimation^{모수추정}
4. Tests of Hypothesis^{가설검정}
5. Confidence Intervals^{신뢰구간}
6. Predictions^{예측}
7. Measuring the Quality of Fit^{적합도측정}

Covariance^{표본공분산} & Correlation Coefficient^{표본상관계수}

They measure both the direction and the strength of the linear relationship between two variables.

There can be many ways of measuring these.

In this class, we will mainly consider Pearson's covariance/correlation coefficient.

Notation for Bivariate Data

Table: Notation for bivariate data

ON	Y	X
1	y_1	x_1
2	y_2	x_2
$\vdots$	$\vdots$	$\vdots$
n	y_n	x_n

Averages 표본평균 of Y and X

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}, \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

ON는 observation number의 약어이다.

Positive Association 양연관성

Figure: 양연관성 산점도

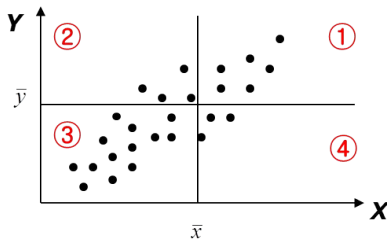


Table: 각 사면quadrant에서의 부호

Quadrant	$y_i - \bar{y}$	$x_i - \bar{x}$	$(y_i - \bar{y})(x_i - \bar{x})$
1	+	+	+
2	+	-	-
3	-	-	+
4	-	+	-

Negative Association 음연관성

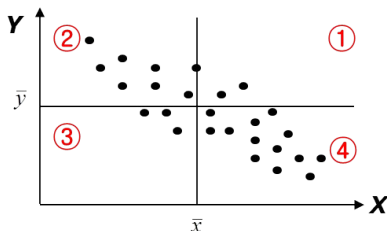


Figure: 음연관성 산점도

Quadrant	$y_i - \bar{y}$	$x_i - \bar{x}$	$(y_i - \bar{y})(x_i - \bar{x})$
1	+	+	+
2	+	-	-
3	-	-	+
4	-	+	-

Table: 각 사면에서의 부호

Definition and Interpretation of Covariance

Definition

$$\text{cov}(Y, X) = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{n - 1}$$

Interpretation:

- ▶ If it is positive, then there is a positive relationship.
- ▶ Otherwise, then there is a negative relationship.

$\text{cov}(Y, X)$ 는 **모집단** 공분산과 구별하기 위해 **표본** 공분산으로 부른다.

Sign of Covariance

Let Q_i be i -th quadrant for $i = 1, \dots, 4$. Consider the plot for the positive association.

$$\begin{aligned} & \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) \\ = & \sum_{(x_i, y_i) \in Q_1, Q_3} (y_i - \bar{y})(x_i - \bar{x}) + \sum_{(x_i, y_i) \in Q_2, Q_4} (y_i - \bar{y})(x_i - \bar{x}) \\ = & (\text{sum of positive numbers}) + (\text{sum of negative numbers}) \\ = & \sum_{(x_i, y_i) \in Q_1, Q_3} |(y_i - \bar{y})(x_i - \bar{x})| - \sum_{(x_i, y_i) \in Q_2, Q_4} |(y_i - \bar{y})(x_i - \bar{x})| \\ > & 0, \end{aligned}$$

since most data points in the first and third quadrants give larger values of $|(y_i - \bar{y})(x_i - \bar{x})|$ than those of the 2nd and 4th quadrants.

One can argue similarly for negative association.

Positive Linear Relationship

Meaning:

- ▶ As X increases, Y increases on the average.

Sign of covariance shows positivity of linear relationship.

- ▶ There are more points in the 1st & 3rd quadrants than in the 2nd & 4th quadrants.
- ▶ The sum of the last column in Table 2 is likely to be positive.

Negative Linear Relationship

Meaning:

- ▶ As X increases, Y decreases on the average.

Sign of covariance shows negativity of linear relationship.

- ▶ There are more points in 2nd & 4th quadrants than in the 1st & 3rd quadrants.
- ▶ The sum of the last column in Table 3 is likely to be negative.

Scaling 척도화, 스케일링 via Standardization 표준화

Some quantities may depend on the units of measurement.

Covariance is one such example.

Whenever the dependence on the unit causes some problems, then we standardize the data.

Definition of Standardization:

$$z_i = \frac{y_i - \bar{y}}{s_y} \quad \text{with} \quad s_y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$$

Here y_i 's are the original data and z_i 's are standardized data.

연관성이 단위에 따라 달라진다고 생각하기는 어렵다. 예컨대, 부자간의 키의 유전 유사성이 키를 측정할 때 cm로 할 때와 inch로 할 때 달라진다는 것은 납득하기 어렵다.

다른 특성을 나타내는 통계량 **statistic**의 척도불변성 **scale invariance**을 위해 표준화를 사용하는 것이 일반적이다.

Definition of Pearson's Correlation Coefficient

Covariance depends on the units of measurement so that it may not be a good measure for association.

If we standardize X and Y , then we can remedy such scale dependence.

Definition Pearson's Correlation Coefficient

$$\begin{aligned}\text{corr}(Y, X) &= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{y_i - \bar{y}}{s_y} \right) \left(\frac{x_i - \bar{x}}{s_x} \right) \\ &= \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum (y_i - \bar{y})^2 \sum (x_i - \bar{x})^2}} \\ &= \frac{\text{cov}(Y, X)}{s_y s_x}\end{aligned}$$

$\text{corr}(Y, X)$ 는 **표본**상관계수라 부르며 모수인 **모집단**상관계수가 별도로 존재하므로 이와 구별된다.

Properties of Pearson's Correlation Coefficient

It does not change if we change the units of measurements.

It is known that $-1 \leq \text{corr}(Y, X) \leq 1$.

Interpretation

- ▶ The closer it is to 1 or -1 , the stronger relationship.
- ▶ $\text{corr}(Y, X) > 0$ implies that Y and X are positively related.
- ▶ $\text{corr}(Y, X) < 0$ implies that Y and X are negatively related.

표본상관계수 절대값이 1 이하라는 사실은 코쉬-슈왈츠 [Cauchy-Schwarz](#) 부등식에 따르는데 그 증명은 생략한다.

Meaning of $\text{corr}(X, Y) = 0$

$\text{corr}(X, Y) = 0$ does not necessarily mean that Y and X are not related.

It only implies that they are not *linearly* related.

For example, Y and X in Figure 3 have the perfect nonlinear relationship $Y = 50 - X^2$, yet $\text{corr}(X, Y) = 0$.

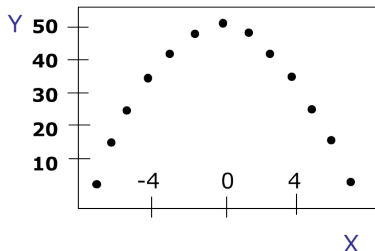


Figure: A scatter plot of Y versus X in Table 2.3

표본상관계수는 선형연관성 [linear association](#)에 대한 척도이며, 여타의 연관성에 대한 정보를 제공하지는 않는다.

Importance of Visualization 시각화

Visualization of data may give more information than that of summary statistic's.

It is important to examine the scatter plot of Y versus X before interpreting the numerical value of the correlation coefficients.

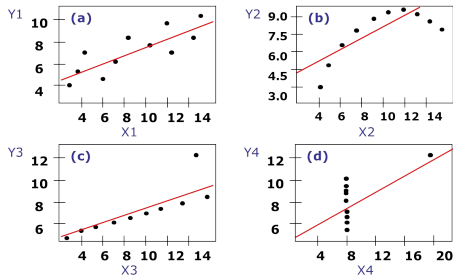


Figure: Scatter plots of the data in Table 2.4 with the fitted lines. (a), (b), (c), (d) give the same value of $\text{corr}(X, Y)$. (Figure 2.3)

시각화는 빅데이터 **big data** 분석에서도 시각화는 대단히 중요한 이슈이다. 특히, 고차원 **high-dimensional** 데이터의 시각화는 간단한 문제가 아니다.

Additional Properties of cov & corr

$\text{cov}(X, Y) = \text{cov}(Y, X)$ and $\text{corr}(X, Y) = \text{corr}(Y, X)$. Hence, there is *no distinction* between two variables in measuring association by covariance and correlation coefficient.

$\text{corr}(X, Y)$ measures only *pairwise* relationships. Regression analysis can be used to relate one or more responses variable to one or more predictors.

Show the following properties for fixed constants a, b, c, d .

1. The standardized variable Z as in (2.3) satisfies $\bar{z} = 0, s_z = 1$.
2. $\text{cov}(Y, X) = \text{cov}(X, Y)$.
3. $\text{cov}(aY + b, cX + d) = ac \text{cov}(Y, X)$.
4. $\text{corr}(aY + b, cX + d) = \text{corr}(Y, X)$.

Computer Repair Data (CRD)

Row	Mininutes	Units	Row	Mininutes	Units
1	23	1	8	97	6
2	29	2	9	109	7
3	49	3	10	119	8
4	64	4	11	149	9
5	74	5	12	145	9
6	87	6	13	154	10
7	96	7	14	166	10

Table: Length of Service Calls (in Minutes) and Number of Units Repaired. Considering the objective of the study, we assume that the response is the length of service calls and the predictor is the number of units. (Table 2.5)

Exploratory Data Analysis for CRD

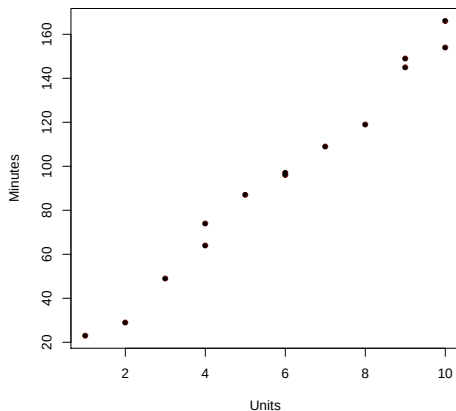


Figure: Computer Repair data: Scatter plot of Units (X) versus Minutes (Y). The scatter plot shows that there is a strong positive linear association between X and Y . (Figure 2.4)

Summary Statistic's for CRD

Scatterplot is shown in Figure 5 which shows that there is a strong linear relationship between X and Y .

Summary statistic's for association between X and Y

- ▶ $\text{cov}(X, Y) = 136$
- ▶ $\text{corr}(X, Y) = 0.996$

The high value of $\text{corr}(X, Y) = 0.996$ is consistent with the scatterplot in in Figure 5 showing strong linear relationship between X and Y .

Now it is time for us to consider a regression model for CRD.

The Simple Linear Regression Model 단순선형회귀모형

Definition

$$Y = \beta_0 + \beta_1 X + \varepsilon \quad (1)$$

X and Y denote the predictor and the response, respectively.

- ▶ CRD: X = Units and Y = Minutes

β_0 and β_1 denote regression coefficients 회귀계수 or parameters.

- ▶ β_1 is called the *slope* 기울기. It may be interpreted as the change in Y for unit change in X .
- ▶ β_0 is called the constant coefficient or *intercept* 절편. It denotes the predicted value of Y when $X = 0$.

ε denotes a random disturbance or *error*.

보통 X 는 고정된 값으로 가정하며 회귀계수들은 모수들이므로 정의상 고정된 값들이다.
따라서 Y 의 랜덤성은 ε 의 랜덤성에 기인한다.

Data from The Simple Linear Regression Model

Observations of X and Y are denoted by lower cases.

Observed data are denoted as in Table 1.

The data set from the model (1) consists of

$$(x_1, y_1), \dots, (x_n, y_n)$$

where

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \text{and} \quad i = 1, \dots, n. \quad (2)$$

Note that we do not actually know ε_i .

보통 확률변수 **random variable**는 대문자로 그 관측값 **observed value**은 소문자로 나타낸다. 그러나 이러한 차이를 아는 사람들은 무시하기도 한다. 한편, X 가 x_i 들로 관측된 후에는 이들 값은 고정된 값으로 간주하며, (2)에서 ε_i 는 더 이상 확률변수가 아닌 관측값이지만 그 값을 알 수는 없다.

A Picture for Data Generation

Since the regression coefficients β_0, β_1 are unknown but *fixed*, the regression line $y = \beta_0 + \beta_1 x$ is unknown but *fixed*.

The dots denote data points (x_i, y_i) satisfying the simple linear regression model (2).

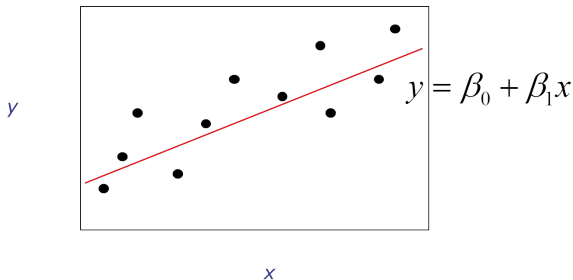


Figure: Dots denote the observed data (x_i, y_i) 's and the red line denotes the unknown regression line.

A Picture for the Relationship between the Population^{모집단} and a Sample^{표본}

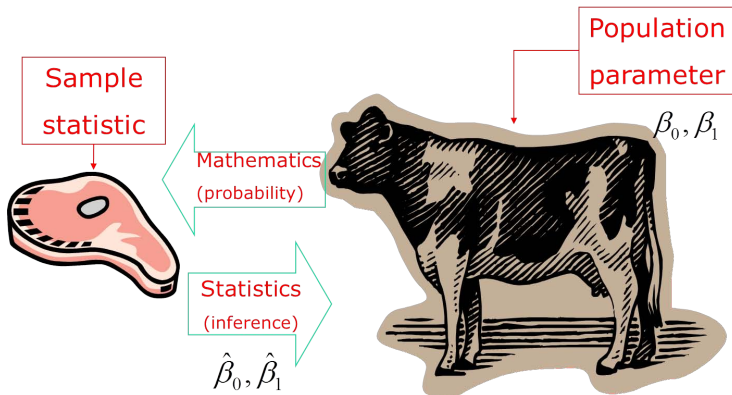


Figure: A picture for the relationship between the population and a sample from the population. A **statistical problem**^{inference 추론} is to guess^{infer} a characteristic (parameter) of the population from a corresponding characteristic (statistic) of sample.

Comparison of Lines

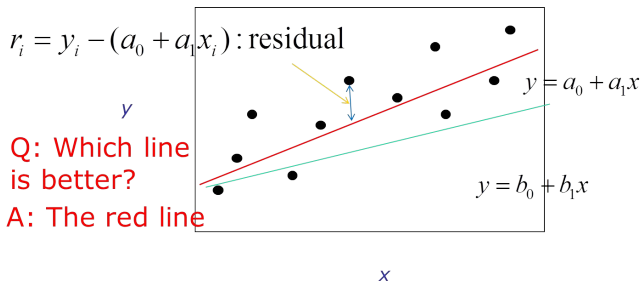


Figure: r_i denotes i -th residual of a line $y = a_0 + a_1 x$. In order to define the *best* line, we need to *order* arbitrary two lines according to one criterion. A popular criterion is the sum of squares of residuals.

A reason for the answer:

- ▶ On the average, the sum of squares of the residuals for the red line is relatively smaller.

Least Squares Estimator 최제추정량 (LSE)

For a vector $b = (b_0, b_1)$, the sum of squares of the residuals is defined as

$$S(b) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

The LSE of β is defined as the vector $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)$ satisfying

$$S(\hat{\beta}) \leq S(b) \text{ for all } b \quad \text{or} \quad \hat{\beta} = \underset{b \in \mathbb{R}^2}{\operatorname{argmin}} S(b).$$

Explicit formula for LSE

$$\hat{\beta}_1 = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

데이터 (x_i, y_i) 들이 관측된 후에는 각 벡터 b 에 대해 하나의 실수 real number $S(b)$ 가 대응하므로 S 는 직선을 규정 specify 하는 벡터 b 의 함수이다.

최제추정량의 구체적인 공식 유도 과정은 본 강의에서는 생략하지만 모든 벡터 b 에 대해 $S(\hat{\beta}) \leq S(b)$ 임을 증명가능하다.

Fitted Values^{적합값} and the Ordinary Least Squares Residuals^{보통최제잔차}

Fitted values

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \quad \text{and} \quad i = 1, 2, \dots, n$$

The ordinary least squares residuals

$$e_i = y_i - \hat{y}_i \quad \text{and} \quad i = 1, 2, \dots, n$$

적합값은 관측된 데이터에 대해 계산된 예측값^{predicted value} $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ 의 특수한 경우이다.

오차 ε_i 는 확률변수인데 반해 보통최제잔차 e_i 는 통계량^{statistic}으로서 모형 적합도를 살펴보는 데에 중요하게 사용된다.

Regression Output for CRD

Variable	Coefficient	s.e.	T-test	P-value
Constant	$\hat{\beta}_0$	s.e.($\hat{\beta}_0$)	t_0	p_0
X	$\hat{\beta}_1$	s.e.($\hat{\beta}_1$)	t_1	p_1

Table: A standard form of a regression output. The meaning of s.e., T-test and P-value will be explained below.

Variable	DF	Parameter Estimate	Standard Error	t-value	$Pr > t $
Intercept	1	4.16165	3.3551	1.24	0.2385
x	1	15.5088	0.50498	30.71	<0.0001

Table: Regression output for CRD. Popular statistical packages such as SAS (`proc reg`) or R (`lm`) provide this regression output.

Fitted values & Residuals for CRD

i	$\hat{y}_i$	e_i	i	$\hat{y}_i$	e_i
1	19.67	3.320	8	97.214	-0.214
2	35.179	-6.179	9	112.723	-3.723
3	50.688	-1.688	10	128.232	-9.723
4	66.197	-2.197	11	143.741	5.259
5	66.197	7.803	12	143.741	1.259
6	81.706	5.294	13	159.249	-5.249
7	97.214	-1.214	14	159.249	6.751

Table: The fitted values ($\hat{y}_i$) and the ordinary least squares residuals (e_i) for CRD

Plotting Regression Line by R

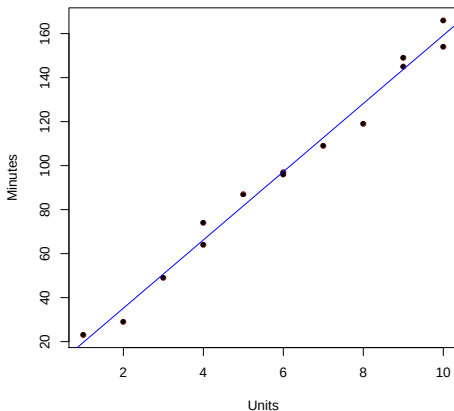


Figure: Plot of Units (X) versus Minutes (Y) with the fitted least squares regression line.

Distribution of LSE

Assumptions on the simple linear regression models

- ▶ The predictor values are fixed.
- ▶ The random errors $\varepsilon_1, \dots, \varepsilon_n$ are distributed independently and identically normal with mean 0 and variance σ^2 .

Under these assumptions, the sampling distributions of LSEs $\hat{\beta}_0$ and $\hat{\beta}_1$ are normal with means β_0 and β_1 and variances

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right] \quad \text{and} \quad \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2},$$

respectively.

여기서 분산은 모집단의 특성이므로 Var로 표현되며 이는 표본분산과 구별된다.

본 강의에서는 최제추정량의 분포성질에 대한 증명은 하지 않지만 이는 증명가능하다.
여기서 $\text{Var}(\hat{\beta}_1)$ 이 x_i 들의 표본분산에 반비례함을 알 수 있다.

Standard Errors of LSEs

For an estimator $\hat{\theta}$ of a parameter θ , an estimator of the standard deviation of $\hat{\theta}$ is called the *standard error* of $\hat{\theta}$ and denoted by $SE(\hat{\theta})$.

An unbiased estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{n-2} = \frac{\sum (y_i - \hat{y}_i)^2}{n-2} = \frac{SSE}{n-2}.$$

Then,

$$SE(\hat{\beta}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2}} \quad \text{and} \quad SE(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}.$$

One may *believe* that the smaller the standard error the more precise the estimator.

표준편차는 모집단의 특성으로 하나의 모수가 되지만 표준오차 *standard error* 는 표준편차의 추정량으로 하나의 추정량이 된다.

A Statistical Reasoning for CRD: Testing 가설검정

Computer Repair Data

- ▶ $\hat{\beta}_1 = 15.5$ & $se(\hat{\beta}_1) = 0.5$
- ▶ Can we say that the population slope β_1 is zero on the basis of these information?

One way of statistical reasoning

- ▶ If we can get LSE of the slope equal to 15.5 *by chance* or quite often under the hypothesis $\beta_1 = 0$, then the difference is not real.
- ▶ If we can get LSE of the slope equal to 15.5 once out of 100 times of collection of such data under the hypothesis, then the *difference* appears to be real.

확률변수는 대문자로 그 관측값은 소문자로 나타내기로 하였지만 β 는 대소문자 구별이 없으므로 한가지 형태로 나타낸다. 한편 표준오차 $SE(\hat{\beta}_1)$ 는 추정량인데 반해, 0.5는 추정치 *estimate*로서 추정량 $SE(\hat{\beta}_1)$ 에 대한 하나의 관측값이므로 소문자 $se(\hat{\beta}_1)$ 로 표현한다.

이러한 방식의 논증 방식을 보통 가설검정 *hypothesis testing* 또는 유의성검증 *test of significance*이라 하는데 본 강의에서는 가설검정 방식을 이용한다.

Key Idea of Testing

The difference should be calculated based on SE.

- ▶ Suppose $SE = 50$. Then 15.5 seems to happen *quite often* under the hypothesis $\beta_1 = 0$.
- ▶ Since $SE = 0.5$ for computer repair data, 15.5 does not seem to happen often under the hypothesis $\beta_1 = 0$ and therefore, *the difference seems to be real*.

가설하에서 흔히 일어나는 사건이 발생한 경우에는 가설이 틀렸다고 의사결정 *decision making, testing* 하지 않지만, 가설하에서 일어나기 어려운 사건이 발생하면 *데이터에 기반하여* 가설이 틀렸다고 의사결정하는 것이 자연스럽다. 이러한 논증방식의 대표분야가 재판 판결과 의료 진찰이다.

Null Hypothesis and Alternative Hypothesis

The *null hypothesis* 귀무가설 (H_0)

- ▶ It expresses the idea that an observed difference is due to chance.
- ▶ To test, the null hypothesis has to be set up as a statistical model (e.g. assumptions of linear models).

The *alternative hypothesis* 대립가설 (H_1)

- ▶ It is an another statement about the statistical model.
- ▶ It says that the difference is real.

특정 연구 *study*의 목적은 새로운 사실을 규명하는 것인데 이는 대립가설에 담겨있게 된다. 따라서, 연구 수행 결과 귀무가설이 맞다고 의사결정, 즉 검정하는 경우는 기존 결과가 틀리다고 결론을 내리지 못하게 되므로 연구 목적을 달성하지 못하게 되었다고 하는 것이 일반적이다.

P-Value

Definition 1 (P-value)

*P-value or the **observed significance level** is the chance of getting a test statistic as extreme as, or more extreme than, the observed one.*

Notes on P-value

- ▶ The chance is computed on the basis that the null hypothesis is right.
- ▶ Extremeness depends on the alternative hypothesis.
- ▶ The smaller this chance is, the stronger the evidence against the null.

유의확률 **P-value** 은 귀무가설과 검정통계량 값의 부합정도를 확률로 표현한 것이다.
따라서, 유의확률로 귀무가설이 맞을 확률을 언급하는 것은 말이 되지 **make sense** 않는다.

Steps of Testing

Steps:

1. Set up the null and the alternative hypotheses
2. Pick a test statistic and set significance level α .
3. Compute the P-value in order to measure the difference between the data and what is expected on the null hypothesis.
4. If P-value is less than α , we conclude that the difference is real i.e. reject the null hypothesis; otherwise accept the null hypothesis.

Old days

- ▶ If P-value is less than $\alpha = 0.05$, the result is called *statistically significant*.
- ▶ If P-value is less than $\alpha = 0.01$, the result is called *highly statistically significant*.

최근에는 유의수준 *significance level* 을 설정하지 않고, 유의확률을 제시한 후 분야 또는 사용자에게 따라 적절히 의사결정에 사용하는 것이 일반적이다. 한편, 최근 유의확률에 대한 여러 비판이 있음에 주목할 필요가 있다.

Testing $\beta_1 = 0$

Null and alternative hypotheses are

$$H_0 : \beta_1 = 0 \quad \text{and} \quad H_1 : \beta_1 \neq 0.$$

Note that H_0 says X is useless as a predictor of Y .

The test statistics

$$t_1 = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)}$$

is distributed as a Student's t distribution *under* H_0 .

When the significance level is α , we reject H_0 if

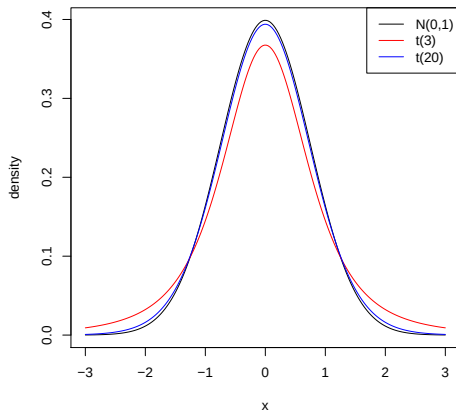
$$|t_1| \geq t_{(n-2, \alpha/2)}$$

or

$$p(|t_1|) \leq \alpha.$$

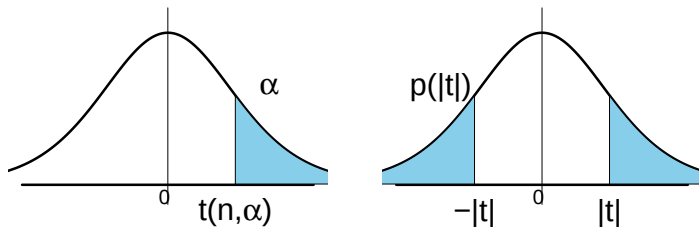
기호 $t_{(n-2, \alpha/2)}$ 와 $p(|t_1|)$ 는 추후 정의된다.

Density Plots for Standard Normal and t Distributions



정규분포 밀도함수가 원점에서 제일 높고 양 꼬리로 가면 다른 t 분포 밀도함수들 보다 그 값이 작게 된다. t 분포 자유도가 클수록 원점에서 상대적으로 밀도함수 값이 높고 양 꼬리로 가면 다른 t 분포 밀도함수들 보다 그 값이 작게 된다.

Two Notations for t distribution with degrees of freedom n



자유도가 n 인 t 분포에 따르는 확률변수를 T 라 하면 T 가 점 $t(n, \alpha)$ 보다 클 확률은 $\mathbb{P}(T > t(n, \alpha)) = \alpha$ 이다.

그림에서 t 분포의 대칭성으로 $p(|t(n, \alpha/2)|) = \alpha$ 임을 알 수 있다.

Test of the Usefulness of the Predictor for CRD

Variable	Coefficient	s.e.	t-test	P-value
Constant	4.162	3.355	1.24	0.2385
Units	15.509	0.505	$30.701 = \frac{15.509}{0.505}$	< 0.0001

Table: Regression Output for the CRD (Table 2.9)

The P-value for $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ is given by $p(30.701) < 0.0001$.

- ▶ Under H_0 , 30.701 does not seem to happen by chance since we obtain such result once out of more than $\frac{1}{0.0001} = 10,000$ times of experiments.
- ▶ The result 30.701 is extremely rare to obtain under H_0 and we have obtained such result.
- ▶ We conclude that the null hypothesis does not seem to be right, i.e. we reject the null hypothesis.

Confidence Intervals^{신뢰구간} for Regression Coefficients and Their Interpretation

Formula

$$\hat{\beta}_0 \pm t_{(n-2, \alpha/2)} \times \text{SE}(\hat{\beta}_0)$$

and

$$\hat{\beta}_1 \pm t_{(n-2, \alpha/2)} \times \text{SE}(\hat{\beta}_1)$$

The confidence interval has the usual interpretation.

- ▶ Suppose we were to take repeated samples of the size at the same values of X and construct for example 95% confidence intervals for the slope parameter for each sample.
- ▶ Then 95% of these intervals would be expected to contain the true value of the slope.

주어진 데이터에 대해 β_1 에 대한 95% 신뢰구간이 (1,2)가 나왔다고 할 때 $P(1 < \beta_1 < 2) = 0.95$ 라 하는 것은 말이 되지 않는다. 이는 확률변수와 그 관측값 간의 관계로 이해할 수 있으며 모수는 고정되어 있고 신뢰구간이 확률변수들로 정해진다.

Why Confidence Intervals hold? I

If the simple linear regression model satisfies the assumptions, then the statistic

$$t_1 = \frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)}$$

has a Student's t -distribution with d.f. $(n - 2)$ *under the assumption that the slope is β_1* . Hence,

$$\mathbb{P}(|t_1| < t_{(n-2, \alpha/2)}) = \mathbb{P}\left(\left|\frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)}\right| \leq t_{(n-2, \alpha/2)}\right) = 1 - \alpha.$$

$\iff$

$$\mathbb{P}\left(|\hat{\beta}_1 - \beta_1| \leq t_{(n-2, \alpha/2)} \text{SE}(\hat{\beta}_1)\right) = 1 - \alpha$$

$\iff$

$$\mathbb{P}\left(\hat{\beta}_1 - t_{(n-2, \alpha/2)} \text{SE}(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{(n-2, \alpha/2)} \text{SE}(\hat{\beta}_1)\right) = 1 - \alpha.$$

Why Confidence Intervals hold? II

Thus a *random* interval

$$\left[\hat{\beta}_1 - t_{(n-2, \alpha/2)} \text{SE}(\hat{\beta}_1), \hat{\beta}_1 + t_{(n-2, \alpha/2)} \text{SE}(\hat{\beta}_1) \right]$$

covers β_1 with probability $1 - \alpha$.

Hence,

$$\hat{\beta}_1 \pm t_{(n-2, \alpha/2)} \text{SE}(\hat{\beta}_1)$$

is an $100(1 - \alpha)\%$ confidence interval for β_1 .

Using a Graphical Method in Measuring the Quality of Fit

One can use a scatterplot for measuring the quality of fit.

- ▶ Scatterplot between Y and X : The closer the set of points to a straight line, the stronger the linear relationship between Y and X .
- ▶ Scatterplot of Y versus the fitted value: The closer the set of points to a straight line, the stronger relationship.

A Plot showing Decomposition of Sum of Squares

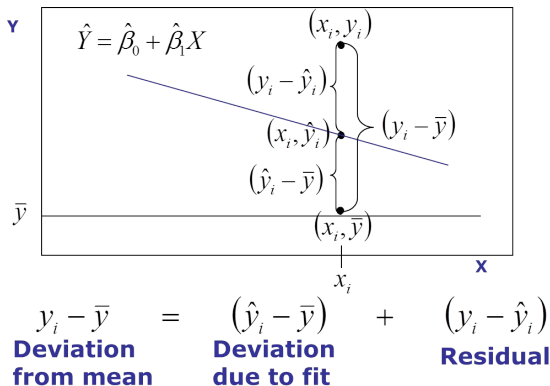


Figure: A graphical illustration of various quantities computed after fitting a regression line to data. (Figure 2.7)

Decomposition of Sum of Squares

One can prove a formula for decomposition of sum of squares

$$SST = SSR + SSE$$

where

$$SST = \sum (y_i - \bar{y})^2$$

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

$$SSE = \sum (y_i - \hat{y}_i)^2.$$

위 제곱합 분해 공식은 단순선형모형 뿐만 아니라 추후 배우게 될 중회귀모형에서도 성립하는데 이에 대한 증명은 본 강의에서는 생략한다.

Coefficient of Determination 결정계수

Definition of coefficient of determination:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Properties of R^2 :

$$R^2 = [\text{corr}(Y, \hat{Y})]^2 = [\text{corr}(Y, X)]^2$$

and

$$0 \leq R^2 \leq 1$$

결정계수의 성질은 제곱합 분해 공식과 마찬가지로 단순선형모형 뿐만 아니라 추후 배우게 될 중회귀모형에서도 성립하는데 이에 대한 증명은 본 강의에서는 생략한다.

Interpretation of Coefficient of Determination R^2

R^2 is a goodness-of-fit index.

- ▶ It can be interpreted as the proportion of the total variation in Y that is accounted for by the predictor variable X .

R^2 is known as the coefficient of determination.

- ▶ If R^2 is near 1, then X accounts for large part of the variation in Y .
- ▶ CRD: $R^2 = 0.987$ indicates that nearly 99% of the total variability in Minutes (response) is accounted for by Units (predictor).

Warnings against R^2

There does not seem to exist a formal distribution theory for R^2 .

For example, $R^2 = 0.6$ does not necessarily mean that a simple linear regression model is useful.

One has to use the t -test for the slope in order to make a formal decision making (testing).

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